

Xiankui Yang, Ph.D.

Postdoctoral Scholar | Department of Mathematics and Statistics

University of South Florida

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EDUCATION

Ph.D. in Mathematics, Concentration in Statistics

August 2026

University of South Florida, Tampa, FL

Dissertation: New Developments in Space-Filling Designs

M.A. in Statistics

December 2020

University of South Florida, Tampa, FL

Thesis: Numerical Study of Gap Distributions in Determinantal Point Process on Low-Dimensional Spheres: L-Ensemble of $O(n)$ Model Type for $n = 2$ and $n = 3$

B.S. in Mathematics and Applied Mathematics

June 2016

Central China Normal University, Wuhan, China

RESEARCH INTERESTS & TECHNICAL SKILLS

Research Interests: Design of Experiments, Optimal Design, Response Surface Methodology, Optimization, Statistical Modeling, Statistical Learning, Machine Learning, Travel Behavior Analysis

Programming & Statistical Computing: R, Python, SQL, SAS, JMP

Software & Tools: Git, PostgreSQL, R Shiny, ArcGIS, LaTeX

PROFESSIONAL EXPERIENCE

Postdoctoral Scholar

August 2026 – Present

Department of Mathematics and Statistics

University of South Florida

- Develop statistical models integrating logistic regression and neural networks, with emphasis on alternative loss functions and estimation methods.
- Conduct research in optimal experimental design, multi-objective optimization, and statistical and machine learning methods.
- Develop computational algorithms in R and Python for statistical design and optimization.

Graduate Teaching Assistant

September 2023 – August 2026

Department of Mathematics and Statistics

University of South Florida

- Assisted with undergraduate and graduate statistics courses through grading assignments and exams, tutoring students, and providing course support.
- Supported courses in probability, mathematical statistics, applied statistics, multivariate statistics, nonparametric statistics, reliability, time series, and introductory statistics.
- Assisted with exam preparation and administration, provided office-hour and individual student support, and maintained accurate grading records.

Graduate Research Assistant

September 2020 – August 2023

School of Public Affairs

University of South Florida

- Analyzed large-scale transportation, mobility, and survey datasets, including Washington State Commute Trip Reduction data, Metropia ride-matching data, and U.S. Census Household Pulse Survey data, to investigate travel behavior, carpooling adoption, mobility disparities, and congestion mitigation.
- Applied statistical modeling, machine learning, survival analysis, simulation, spatial analysis, and data visualization methods to evaluate transportation programs, identify behavioral and demographic factors, and support policy and planning analyses.
- Contributed to the full research workflow, including research design, data cleaning and processing, statistical analysis and model development, results interpretation and visualization, technical report preparation, and manuscript writing.

PUBLICATIONS

Peer-Reviewed Journal Articles

1. **Yang, X.**, Lu, L., & Anderson-Cook, C. M. (2026). Quick Input-Response Space-Filling (QIRSF) Designs. *Quality and Reliability Engineering International*. <https://doi.org/10.1002/qre.70062>
2. Chen, P., **Yang, X.**, Hao, F., & Tao, R. (2026). Decoding electric vehicle adoption using XGBoost and SHAP analysis. *Travel Behaviour and Society*, 42, 101125. <https://doi.org/10.1016/j.tbs.2025.101125>
3. Chen, P., & **Yang, X.** (2025). Premature casual carpooling in Texas: analyzing customer churn in the Metropia experiment with survival analysis and machine learning. *Case Studies on Transport Policy*, 101587. <https://doi.org/10.1016/j.cstp.2025.101587>
4. **Yang, X.**, Lu, L., & Anderson-Cook, C. M. (2025). A scalable algorithm for generating non-uniform space-filling designs. *Journal of Quality Technology*, 1–19. <https://doi.org/10.1080/00224065.2025.2513504>
5. Chen, P., **Yang, X.**, Zhang, Y., & Hud, S. (2025). The resiliency of transit and ride-hailing system through the coronavirus pandemic: Lessons from six large US cities. *Case Studies on Transport Policy*, 101470. <https://doi.org/10.1016/j.cstp.2025.101470>
6. Tao, R., **Yang, X.**, Hao, F., & Chen, P. (2024). Demographic disparity and influences in electric vehicle adoption: A Florida case study. *Transportation Research Part D: Transport and Environment*, 136, 104465. <https://doi.org/10.1016/j.trd.2024.104465>
7. Chen, P., **Yang, X.**, & Zhang, Y. (2024). Exploring the interconnection between telecommuting, trip reduction, and transportation equity amidst the coronavirus pandemic. *Transportation Letters*, 1–11. <https://doi.org/10.1080/19427867.2024.2366377>
8. **Yang, X.**, Lu, L., & Anderson-Cook, C. M. (2024). Input-response space-filling designs incorporating response uncertainty. *Statistical Analysis and Data Mining: The ASA Data Science Journal*, 17(1), e11648. <https://doi.org/10.1002/sam.11648>
9. **Yang, X.**, Chen, P., & Zhang, Y. (2023). How mentally healthy we are during the pandemic? findings from changed travel behavior. *Journal of Transport & Health*, 29, 101587. <https://doi.org/10.1016/j.jth.2023.101587>
10. Chen, P., & **Yang, X.** (2023). Revisit employer-based travel demand management: A longitudinal analysis. *Transport Policy*, 131, 22–31. <https://doi.org/10.1016/j.tranpol.2022.11.014>
11. Chen, P., Zhai, W., & **Yang, X.** (2023). Enhancing resilience and mobility services for vulnerable groups facing extreme weather: lessons learned from Snowstorm Uri in Harris County, Texas. *Natural Hazards*, 118(2), 1573–1594. <https://doi.org/10.1007/s11069-023-06062-2>

Technical Reports

1. Rodriguez-Roman, D. A., Chen, P., **Yang, X.**, Carlo Colón, H., Santiago, N. G., Tao, R., & Figueroa Medina, A. M. (2024). *Carpool-based Parking Assignment Policy*. National Institute for Congestion Reduction. <https://doi.org/10.5038/CUTR-NICR-Y3-2-10>
2. Chen, P., Nguyen, T., Winters, P., **Yang, X.**, Lee, D. H., Li, W., & Quadrifoglio, L. (2023). *Social Carpooling-Based Road Congestion Mitigation: A Three-Level Analysis*. National Institute for Congestion Reduction. https://digitalcommons.usf.edu/cutr_nicr/33/
3. Chen, P., **Yang, X.**, Winters, P. L., Li, M., Rodriguez, D. A., Montilla, M. A. N., & Chatman, D. G. (2021). *Impacts of commute trip reduction programs, rail station area built environment changes, and ride-hailing services on traveler behavior*. National Institute for Congestion Reduction. https://digitalcommons.usf.edu/cutr_nicr/12/
4. Chen, P., **Yang, X.**, & Zhang, Y. (Submitted). *Mobility Challenges of Female Minority Workforce During the Coronavirus Pandemic*. Center for Transportation, Environment, and Community Health

SOFTWARE & APPLICATIONS

1. QNUSF — Quick Non-Uniform Space-Filling Designs
R package and interactive R Shiny application for generating non-uniform space-filling designs.
R Package: [GitHub](#) | **Web App:** [R Shiny](#)
2. QIRSF — Quick Input-Response Space-Filling Designs
R package and interactive R Shiny application for generating input-response space-filling designs.
R Package: [GitHub](#) | **Web App:** [R Shiny](#)

CONFERENCE PRESENTATIONS

1. ASQ and ASA Fall Technical Conference 2025, Invited Talk. “A scalable algorithm for generating non-uniform space-filling designs.” October 7–9, 2025, Houston, Texas.
2. ASQ and ASA Fall Technical Conference 2024, Contributed Talk. “Quick Input-Response Space Filling (QIRSF) Designs.” October 9–10, 2024, Nashville, Tennessee.
3. Florida Planning Conference 2024, Poster Presentation. “EV Adoption in Florida: Progresses and Challenges.” September 3–6, 2024, Tampa, Florida.
4. 2024 Joint Statistical Meetings, Contributed Talk & Poster Presentation. “Understanding Post-Pandemic Mental Health via Statistical Learning.” August 3–8, 2024, Portland, Oregon.
5. Joint Research Conference 2024, Contributed Talk. “Quick Non-Uniform Space-Filling (QNUSF) Designs.” June 17–20, 2024, Waterloo, Ontario, Canada.
6. Florida Chapter of the ASA 2024 Meeting, Contributed Talk. “Input-response space-filling designs incorporating response uncertainty.” March 29–30, 2024, Tallahassee, Florida.
7. Transportation Research Board 103rd Annual Meeting. January 7–11, 2024, Washington, DC
 - Poster Presentation: “Peer to Peer Carpooling: Who Carpool and How Often Do People Carpool?”
 - Poster Presentation: “Trip Plan App for Peak Avoidance and Casual Carpooling: A Focus on Customer Churn.”
 - Contributed Talk: “Casual Carpooling with Incentivized Parking at Universities.”
8. Transportation Research Board 102nd Annual Meeting, Poster Presentation. “The renaissance of transit and ridesharing: from pandemic towards the new normal.” January 8–12, 2023, Washington, DC

AWARDS & FUNDING

1. **Tharp Endowed Award**, Fred L. and Helen M. Tharp, 2026 — \$1,306
2. **Student Travel Award**, FTC, 2025 — \$800
3. **Student Travel Award**, Department of Mathematics and Statistics, University of South Florida, 2025 — \$400
4. **Tharp Endowed Award**, Fred L. and Helen M. Tharp, 2025 — \$1,500
5. **Student Travel Award**, Department of Mathematics and Statistics, University of South Florida, 2024 — \$400
6. **Student Travel Award**, Virginia Tech, 2024 — \$1,000
7. **Tharp Endowed Award**, Fred L. and Helen M. Tharp, 2024 — \$1,250
8. **Student Travel Award**, Center for Urban Transportation Research (CUTR), University of South Florida, 2024 — \$1,733
9. **Student Travel Award**, Center for Urban Transportation Research (CUTR), University of South Florida, 2023 — \$2,257

TEACHING EXPERIENCE

Instructor

- **STA 4222 – Sample Survey Design**, Spring 2026
- **STA 4102 – Methods for Applied Statistics**, Fall 2024

Teaching Assistant

- **Summer 2026**: STA 2023 Introductory Statistics I
- **Fall 2025**: STA 5446 Probability Theory I; STA 5166 Statistical Methods I; STA 4702 Multivariate Statistical Methods
- **Spring 2025**: STA 4322 Introduction to Mathematical Statistics II; STA 4502 Nonparametric Statistical Methods; STA 4671 Reliability
- **Summer 2024**: STA 2023 Introductory Statistics I
- **Spring 2024**: STA 3024 Introductory Statistics II; STA 4442 Introduction to Probability; Smart Lab
- **Fall 2023**: MTG 3207 Geometry Connections; MTG 3212 Geometry; STA 4852 Applied Time Series; Smart Lab

PROJECT EXPERIENCE

NICR Year 1: Targeting Transit Incentives to Congestion Sources

- Analyzed Washington State Commute Trip Reduction (CTR) longitudinal data to evaluate how employer-provided transit subsidies, parking policies, and other travel demand management strategies influence commuting behavior.
- Applied statistical modeling and longitudinal data analysis to examine changes in transit use and commuting patterns and assess the effectiveness of transportation incentives.
- Contributed to data preparation, results interpretation, technical reporting, and manuscript preparation for research on transportation demand management and congestion reduction.

NICR Year 2: Social Carpooling-Based Road Congestion Mitigation: A Three-Level Analysis

- Analyzed Metropia ride-matching data to identify behavioral and program-related factors associated with participation in peer-to-peer social carpooling.
- Developed statistical and simulation analyses to evaluate incentive effectiveness, estimate potential increases in carpool trips across traffic zones, and assess their implications for congestion reduction.
- Contributed to research design, data analysis and modeling, interpretation of results, technical reporting, and manuscript preparation in the areas of shared mobility and travel behavior.

CTECH Year 6: Mobility Challenges of Female Minority Workforce During the Coronavirus Pandemic

- Analyzed U.S. Census Household Pulse Survey data to investigate pandemic-related changes in telework, transit use, mobility, and economic conditions across demographic groups.
- Conducted statistical modeling of sociodemographic disparities to evaluate differences in mobility and access to transportation resources during the COVID-19 pandemic.
- Contributed to project organization, data analysis, results interpretation and reporting, and manuscript preparation, supporting research on transportation equity and workforce mobility.

NICR Year 3: Carpool-Based Parking Assignment Policy

- Analyzed data supporting a carpool-based parking reservation system that integrates ride-matching and parking allocation to improve parking utilization and reduce congestion.
- Conducted statistical analysis and modeling to evaluate public perceptions and the feasibility of using parking incentives to encourage carpool participation.
- Contributed to modeling, results interpretation, technical reporting, and manuscript preparation for the proof-of-concept implementation at the University of Puerto Rico at Mayagüez.

USF Interdisciplinary Research Grant: EV Adoption & Charging Infrastructure in Florida

- Co-organized the research project and contributed to questionnaire design and survey development to investigate EV adoption, charging access, and public perceptions in Florida.
- Integrated EV ownership, charging-infrastructure, GIS, and survey data to examine spatial and demographic disparities between EV demand and public charging availability.
- Conducted statistical and spatial analyses, interpreted results, and contributed to reporting and manuscript preparation to support research on EV adoption and equitable charging-infrastructure planning.

PEER-REVIEW SERVICE

- IISE Transactions
- Transportation Research Part D: Transport and Environment
- Quality Engineering
- Computational Statistics and Data Analysis
- Journal of Transport and Land Use
- Employee Relations
- The Annals of Applied Statistics
- Transportation Research Record
- Travel Behaviour and Society
- Journal of Transport & Health
- Transportation
- Case Studies on Transport Policy